DS	**Session**	Searching	**Question Information**	Level 1 Challenge 10

Question Description:
An array A contains integers with the following constraints:

'A' contains elements in sorted order.
Integer i occurs i*floor(sqrt(i))+ceil(i/2) times in the array.
All elements are greater than or equal to 1.

Fathima has given Q queries of type:

L R: Find the number of distinct values present in subarray A[L...R].
Note: 1-based indexing is followed.

Constraints:
1 <= Q <= 10^5
1 <= L <= R <= 10^13

Input Format:
The first line contains an integer Q denoting the number of queries.
Next Q lines contains two space-separated integers L, R, denoting the query.

Output Format:
Print the output in separate lines contains required number of distinct values.

Example

Sample Input:

2

1 3

1 6

Sample Output:

2

3

Explanation

First few elements of the array A are $1, 1, 2, 2, 2, 3, 3, 3, 3, 3, \ldots$

- For Query 1:-
  - Number of distinct elements in subarray $A[1...3]$ is 2.
- For Query 2:-
  - Number of distinct elements in subarray $A[1...6]$ is 3.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
3 3
2 6
```

EXPECTED OUTPUT

```
1
3
```

Test Case 2

INPUT (STDIN)

```
3
3 3
2 6
5 4
```

EXPECTED OUTPUT

```
1
3
1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(l<ans1)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
6
```

Test Case 2

TOKEN COUNT

```
400
```

Test Case 3

NLOC

```
56
```

Code

solution.c

```c
#include <stdio.h>
#include <math.h>

long long getCount(long long i) {
    return i * (long long)floor(sqrt(i)) + (i + 1) / 2;
}

long long getElementAt(long long idx) {
    long long low = 1, high = 1000000, ans = 1;
    while (low <= high) {
        long long mid = low + (high - low) / 2;
        long long total = 0;
        for (long long k = 1; k <= mid; k++) total += getCount(k);
        if (total >= idx) { ans = mid; high = mid - 1; }
        else low = mid + 1;
    }
    return ans;
}

int main() {
    int q;
    if (scanf("%d", &q) != 1) return 0;
    while (q--) {
        long long l, r;
        scanf("%lld %lld", &l, &r);
        long long valL = getElementAt(l);
        long long valR = getElementAt(r);
        long long distinct = valR - valL + 1;
        printf("%lld\n", distinct);
    }
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
2
1 6
2 6
3
2
PS C:\C Programming>
```